# Chapter 13 In-Class Exercise: ComplexNumber Inheritance

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## 1 Parent Class

Assume `vector2d.py` contains:

```
import math
class Vector2D:
    def __init__(self, x, y):
        self.x = x
        self.y = y
    def magnitude(self):
        return math.sqrt(self.x**2+self.y**2)
    def angle(self):
        return math.atan2(self.y,self.x)
    def __add__(self, other):
        return Vector2D(
            self.x+other.x,self.y+other.y)
    def __sub__(self, other):
        return Vector2D(
            self.x-other.x,self.y-other.y)
```

A complex number $z = a + jb$ also has two components. We will use $x \leftrightarrow a, \qquad y \leftrightarrow b$.

## 2 Create the Child Class

Create `complex_number.py`:

```
from vector2d import Vector2D
class ComplexNumber(Vector2D):
    def __init__(self, real, imag):
        super().__init__(____,____)
```

The expression `ComplexNumber(Vector2D)` means that `ComplexNumber` **inherits** from `Vector2D`. Test:

```
z1 = ComplexNumber(3, 4)
print(z1.magnitude())
print(z1.angle())
```

## 3 Override Addition

Now try

```
z1 = ComplexNumber(3,2)
z2 = ComplexNumber(5,-4)
z3 = z1 + z2
print(type(z3))
```

The arithmetic is correct, but the inherited method returns a `Vector2D`. Override it:

```
def __add__(self, other):
    return ComplexNumber(
        self.x + other.x,
        self.y + other.y)
```

What should `type(z3)` give, if you recalculate $z1+z2$? Override the subtraction method in the same way.

## 4 New Behavior: Multiplication

Complex multiplication is

$$(a + jb)(c + jd) = (ac - bd) + (ad + bc)j.$$

Add:

```
def __mul__(self, other):
    real = ____
    imag = ____
    return ComplexNumber(real,imag)
```

Test the new method.

## 5 Complex-Number Names

The parent stores `x` and `y`. Add:

```
def real(self):
    return self.x
def imag(self):
    return self.y
```

Then test

```
print(z1.real())
print(z1.imag())
```

**Key idea:** Inheritance reuses behavior; overriding replaces inherited behavior when the child class needs something different.